

Use-Case Flow Specification

Problem Statement #27 – Warehouse Inventory & Pallet Location Tracker

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Use Case ID: UC-02

Use Case Name: Store pallet in bin

Primary Actor: Warehouse Operator

Secondary Actor: Barcode / RFID Scanner (device)

Related Requirements: FR-001, FR-003, FR-004

Relationships: «include» UC-03 Validate rack load capacity; «include» UC-06 Log stock movement; extended by UC-04 Suggest alternative bin

Trigger: A registered pallet is on the forklift at the inbound dock and the operator requests a put-away.

Preconditions

1. The operator is authenticated and holds the Warehouse Operator role.
2. The pallet has already been registered through UC-01, so its SKU, quantity and gross weight are on record.
3. At least one bin in the rack layout is flagged vacant.
4. Rack master data — rated load capacity and current total load — is available to the system.

Postconditions

On success: the pallet record holds a unique aisle-rack-level coordinate; the target bin is flagged occupied; the rack's running total load is increased by the pallet's gross weight and remains within its rated capacity; and a movement log entry has been written.

On failure: the pallet remains unassigned at the inbound dock, no bin is flagged occupied, the rack's total load is unchanged, and the rejected attempt is recorded.

Main Success Scenario

1. The operator selects "Store Pallet" on the handheld terminal.
2. The system prompts for the pallet tag to be scanned.
3. The operator scans the pallet's barcode/RFID tag; the scanner transmits the tag ID to the system.
4. The system retrieves the pallet record and displays its SKU, quantity and gross weight.
5. The system displays the vacant bins together with each rack's remaining spare capacity.
6. The operator selects a proposed bin coordinate (aisle, rack, level).
7. The system validates the proposed placement against the rack's rated load capacity. «include UC-03»
8. Validation passes; the system reserves the bin and asks the operator to confirm the placement.
9. The operator confirms.
10. The system assigns the coordinate to the pallet record and flags the bin occupied.
11. The system writes a movement log entry containing pallet ID, source, destination bin, operator ID and timestamp. «include UC-06»
12. The system displays a confirmation showing the final coordinate. The use case ends successfully.

Alternate Flow 7a — Rack capacity would be exceeded («extend» UC-04 Suggest alternative bin)

- 7a1. At step 7 the system computes that the resulting total load would exceed the rack's rated capacity, and rejects the proposed coordinate.
- 7a2. The system displays a rejection message stating the rack's rated capacity, its current load and the pallet's gross weight.
- 7a3. The system invokes UC-04 and offers up to three alternative vacant bins whose remaining capacity can carry the pallet.
- 7a4. The operator selects one of the alternatives; the flow resumes at step 7 so the new coordinate is validated in the same way.
- 7a5. If no alternative exists, or the operator declines all of them, the system records the failed attempt, leaves the pallet unassigned at the inbound dock and raises a capacity alert to the Logistics Supervisor. The use case ends in failure.

Alternate Flow 3a — Pallet tag unreadable

- 3a1. The scanner fails to read the tag, or returns an ID with no matching pallet record.
- 3a2. The system displays a "tag not recognised" message and prompts for a rescan.
- 3a3. After three failed attempts the system allows the operator to key in the pallet ID manually, and flags the record for supervisor verification. The flow then resumes at step 4.